Oxygenation, Fluid Status, and Immune-Risk Assessment

NURSING · nursing-foundations · BoardPrep lesson v1

Official examination scope

PRC Resolution No. 10 s. 2025 and the August 2026 NLE program define examination scope, not topic weights. This lesson teaches a bounded oxygenation, fluid status, and immune-risk assessment foundation from the references below. Open RN and U.S. professional materials support general educational principles only; apply Philippine law, scope of practice, current clinical orders, referral pathways, and facility policy. This study material is not a clinical directive or proof of complete NP coverage.

Learning objectives

- Use a focused airway and breathing assessment to recognize deterioration, verify relevant
- observations, and escalate under current orders and local pathways.
- Integrate intake/output, perfusion, symptoms, and trends when assessing possible fluid
- imbalance without prescribing an unassessed fluid plan.
- Recognize that immunosuppression can change risk or presentation and that new symptoms warrant
- prompt assessment.

1. Read oxygenation as a changing clinical picture

Assess airway patency, breathing effort and pattern, ability to speak, skin and mental-status change, relevant measured trends, and the patient's baseline. Verify a surprising reading and the sensor or measurement context when this can be done without delaying a safety response. New restlessness with worsening breathing or a downward oxygenation trend is a deterioration cue: position and support the patient within role, provide only currently prescribed oxygen, call for help under the facility pathway, and reassess response. Do not use a single number, device setting, or diagnosis as a substitute for the whole assessment. Numeric thresholds and U.S.-specific oxygen-order rules are intentionally omitted.

Sources for this section

Section reference: 8.2 Oxygenation Basic Concepts — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/8-2-basic-oxygenation-concepts/
Section reference: Basic Emergency Care: Approach to the Acutely Ill and Injured — World Health Organization and International Committee of the Red Cross — https://www.who.int/publications-detail-redirect/9789241513081

2. Combine symptoms and trends in fluid assessment

Fluid status is assessed from more than one cue: recent losses or intake, dizziness or weakness, perfusion and mental-status changes, urine-output trend, weight trend, examination, and ordered laboratory results. Ask about vomiting or other losses and compare findings over time; a single sign is not enough to identify cause or prescribe treatment. A cluster of vomiting, orthostatic dizziness, and markedly reduced output warrants prompt focused assessment and communication with the care team. Do not recommend unlimited fluids or independently change a prescribed restriction. Interpret measurements in context and follow local orders and escalation policy.

Sources for this section

Section reference: 15.2 Basic Fluid and Electrolyte Concepts — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/15-2-basic-fluid-and-electrolyte-concepts/
Section reference: Nursing Fundamentals — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/

3. Treat new symptoms in immune-risk context

Immune suppression may increase infection risk and can alter the expected presentation; absence of a dramatic fever does not rule out infection or another serious problem. Review the known treatment and baseline within role, ask about new symptoms and exposure, assess for deterioration, and report the change promptly through the current clinical pathway. Do not wait for a self-selected fever threshold, administer a live vaccine without qualified review, or infer a cause from immune status alone. The section teaches recognition and escalation, not an infection diagnosis, antimicrobial selection, or local isolation order.

Sources for this section

Section reference: 9.7 Applying the Nursing Process — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/9-7-applying-the-nursing-process/
Section reference: Core Infection Prevention and Control Practices — U.S. Centers for Disease Control and Prevention — https://www.cdc.gov/infection-control/hcp/core-practices/
Section reference: Basic Emergency Care: Approach to the Acutely Ill and Injured — World Health Organization and International Committee of the Red Cross — https://www.who.int/publications-detail-redirect/9789241513081

Respond to worsening breathing

Question: A patient with pneumonia becomes restless, breathes faster, and has a falling oxygenation reading.
1. Assess airway and breathing, work of breathing, mental status, and the observed trend.
2. Verify measurement context if safe; position and provide only prescribed support within role.
3. Escalate through the facility pathway and reassess the response.
Answer: Treat the change as possible deterioration and seek timely help.
Check: No oxygen threshold, device setting, or independent order is supplied.

Assess fluid-loss cues

Question: A patient reports vomiting and dizziness and has a decreasing urine-output trend.
1. Ask about the timing and amount of losses and compare available intake/output and weight trends.
2. Assess perfusion, hydration cues, mental status, and relevant ordered results.
3. Report the pattern and response; follow the current care plan rather than prescribing unrestricted fluids.
Answer: Integrate the pattern and escalate concern; do not rely on one sign.
Check: Arithmetic of fluid totals does not independently establish cause or treatment.

Consider immune-risk context

Question: A person receiving immune-suppressing therapy reports a new fever.
1. Recognize increased risk and ask about onset, associated changes, and relevant treatment context.
2. Assess for deterioration and document objective findings.
3. Promptly notify the responsible team under the local pathway and reassess.
Answer: A new symptom in immune-risk context deserves prompt assessment and communication.

Check: Do not wait for a higher self-selected threshold or assume immune suppression prevents fever.

Common misconceptions

- A falling measurement is not interpreted without checking context, baseline, symptoms, and
- trend.
- A normal-looking single sign does not cancel worsening breathing or perfusion cues.
- Vomiting and low output do not justify unlimited fluids or an independent treatment change.
- Immunosuppression can alter risk or presentation; the absence of a high fever does not prove
- safety.
- A clinical mnemonic supports assessment but does not replace escalation or reassessment.

Practice patterns

- For respiratory change, assess the patient and trend, verify when safe, support within orders,
- escalate, and reassess.
- For fluid concerns, combine loss history, output, perfusion, mental status, and ordered data.
- For immune-risk cases, report new changes promptly without waiting for a numeric threshold.

Official scope references

Official scope: Enhanced Tables of Specifications for the Subjects in the Nurses Licensure Examination — Professional Regulation Commission; Professional Regulatory Board of Nursing — https://www.prc.gov.ph/sites/default/files/2025-10%20%20published%20Enhanced%20TOS%20nuse.pdf
Official scope: Nursing Licensure Examination Program, August 2026 — Professional Regulation Commission; Professional Regulatory Board of Nursing — https://www.prc.gov.ph/sites/default/files/exam%20progam%20aug%202026%20nle%20(nursing)%20signed.pdf

Instructional references

Instructional reference: 8.2 Oxygenation Basic Concepts — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/8-2-basic-oxygenation-concepts/
Instructional reference: 15.2 Basic Fluid and Electrolyte Concepts — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/15-2-basic-fluid-and-electrolyte-concepts/
Instructional reference: 9.7 Applying the Nursing Process — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/9-7-applying-the-nursing-process/
Instructional reference: Basic Emergency Care: Approach to the Acutely Ill and Injured — World Health Organization and International Committee of the Red Cross — https://www.who.int/publications-detail-redirect/9789241513081
Instructional reference: Nursing Fundamentals — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/
Instructional reference: Core Infection Prevention and Control Practices — U.S. Centers for Disease Control and Prevention — https://www.cdc.gov/infection-control/hcp/core-practices/

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